

Distillation, Failure Modes, and Benchmark Fragility in Forensic White-Light Bruise Segmentation:

A Comprehensive Report on the Final Five-Model Run

Injury Analytics Lab — Bruise Segmentation Project

Generated from `results_final/` and the val-selected best-seed re-inference
(final run: `bruise_colab_final.ipynb`; analysis: `bruise_colab_final_analysis.ipynb`)

Abstract

We report a controlled comparison of five semantic-segmentation models for white-light (WL) bruise segmentation on a multi-annotator forensic dataset: a SegFormer-B2 teacher, a SegFormer-B0 student trained directly, a SegFormer-B0 student trained with calibrated soft-target distillation, and two YOLO26n-sem models (direct and distilled) trained with their native Ultralytics recipe. All models are trained only on one annotator’s masks and evaluated against a three-annotator *majority-vote* target on a fixed test set of **185 images from 28 subjects**. The central quantitative facts are: (i) at 28 subjects the minimum resolvable difference in mean Dice is ≈ 0.04 , which is *larger* than the distillation effect (+0.002, n.s.) and the student–teacher gap (−0.001, n.s.); (ii) the one axis that separates the architectures by more than label noise is the *complete-miss rate* (SegFormer 0–0.5% vs. YOLO 2–8%); (iii) the three human experts agree with one another at only **0.639** mean Dice, and every model beats that ceiling—indeed a model trained on a single annotator (Paul) agrees with the three-expert consensus *significantly better than Paul himself does* (+0.07 Dice, $P(\Delta > 0) \approx 0.98$); and (iv) an apparent skin-tone performance gap is confounded with lesion size, though a residual light-skin recall deficit for YOLO survives adjustment for size and subject clustering at the boundary of significance ($p \approx 0.05$). We frame the study as a case study in benchmark fragility: the conclusions turn far more on annotation noise, sample size, and evaluation-harness choices than on architecture. Two external baselines (U-Net and DeepLabV3+, ResNet50) trained through the identical recipe land at 0.753 ± 0.015 and 0.745 ± 0.018 mean Dice—within ≈ 0.02 of the SegFormer models and well inside the annotation-noise band—so with seven architectures now clustering at 0.745–0.765, the label-limited reading only strengthens (Section 13; nnU-Net still pending).

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1 Introduction

Bruises are central evidence in forensic assessment of physical abuse, yet their boundaries are diffuse and their appearance varies with skin tone, lighting, age of injury, and body site. Automated segmentation could standardise documentation, but it inherits two hard problems: (1) the “ground truth” is itself a matter of expert opinion, and expert opinions disagree; and (2) the population for

which three-expert consensus labels exist is small. This report documents a five-model run built specifically to be *honest* about both problems—to measure them rather than to hide them behind a single accuracy number.

We compare a transformer family (SegFormer) against a modern one-stage detector family (YOLO26n-sem), including knowledge distillation from a large teacher to a small student in each family. The comparison is deliberately conservative: every difference is quantified with a *subject-level cluster bootstrap*, and the headline results are the ones that survive it.

2 Dataset and Experimental Design

2.1 A multi-annotator intersection study

The data are white-light photographs from the NIJ bruise dataset, labelled by multiple annotators. The three highest-volume annotators—**Paul**, **Gbarimah**, and **Erik**—were identified. The *strict intersection* of all three (every image labelled by all three annotators) is **185 white-light images from 28 subjects**, and this intersection *is* the fixed test set. The prediction target is the per-pixel **2-of-3 majority vote** (a pixel is bruise if at least two of the three annotators marked it), *not* any single annotator’s mask.

2.2 Train/test discipline

A prior five-fold cross-annotator study selected Paul as the most stable annotator, so all five models are **trained only on Paul’s masks** and **evaluated against the three-annotator majority**. Training data are Paul’s *non-intersection* subjects (831 images / 115 subjects), split **697 train / 134 validation** at the subject level. Because the 28 test subjects are structurally removed from the training pool, subject disjointness is guaranteed by construction (verified: 0 overlapping subjects). This is a *cross-annotator generalisation* test, not an ordinary split.

2.3 $n = 28$ is a ceiling, not a choice

There are 28 test subjects because that is how many subjects *all three* annotators labelled; a 29th cannot be obtained without new annotation. Every “underpowered” statement below should therefore be read as a fact about the field—at the *maximum sample size for which three-annotator consensus labels exist, model-level differences are largely unresolvable*—not as a limitation that more effort could remove.

2.4 Skin-tone grouping

For subgroup analysis we use the Individual Typology Angle (ITA), an objective skin-tone measure computed from image pixels, binned into five Fitzpatrick-like groups (Light II–III, Intermediate III–IV, Tan IV, Brown V, Dark VI). Group sizes range from 24 to 55 images but only ≈ 9 –17 *subjects*, which governs the (weak) statistical power of every fairness statement in Section 9.

3 Architectures

All five models emit a *single* bruise logit at full input resolution, so every downstream component (loss, threshold sweep, metric, benchmark) is architecture-blind.

3.1 SegFormer (B2 teacher, B0 students)

HuggingFace SegFormer with a MiT (Mix-Transformer) hierarchical encoder and a lightweight all-MLP decode head, rebuilt with a 1-class head. Input is ImageNet-normalised inside the model wrapper. The teacher is **MiT-B2** (27 347 393.000 parameters, i.e. 27.35 M); both students are **MiT-B0** (3.71 M). The B0 decode head is randomly initialised on a pretrained backbone.

3.2 YOLO26n-sem

Ultralytics YOLO26n semantic-segmentation model, rebuilt with an $nc=1$ head so it emits one bruise logit directly (structurally mirroring SegFormer’s 1-channel head). `.load()` transfers 360/364 backbone tensors; only the head is new. Input scale is /255 (no ImageNet normalisation), because Ultralytics’ BatchNorms carry running statistics for that distribution—feeding it ImageNet-normalised pixels caps it at Dice 0.479. Parameter count 1.63 M. A structural note relevant to the results: YOLO’s main head operates at stride 8 (80×80 for a 640 input) against SegFormer’s stride 4 (160×160), a $4 \times$ coarser spatial grid that is an architectural ceiling on boundary precision, not a training artefact.

4 Training and Evaluation Protocol

4.1 Shared recipe (SegFormer)

Loss is per-image Dice+BCE (Dice computed per image then averaged, to match the reported metric and to keep gradient on the $\approx 4.7\%$ of pixels that are bruise). Optimiser is AdamW with an encoder/head learning-rate split (6×10^{-5} / 6×10^{-4}), no weight decay on norms and biases, linear warmup then polynomial decay, AMP, and gradient clipping. **Model selection is on a threshold-free binned average precision (AP)** over validation pixels, not on Dice at a fixed 0.5 threshold (the operating threshold is fitted afterwards and is nowhere near 0.5). After training, the decision threshold is **swept on validation and applied once to test**.

4.2 Distillation (what it is, and is not)

The distilled models use *calibrated soft-target* distillation:

$$\mathcal{L} = \alpha \text{DiceBCE}(z_s, y) + (1 - \alpha) \text{BCE}(z_s, \sigma(z_t/T_{\text{cal}})),$$

where T_{cal} is the temperature fitted by minimising validation NLL (Guo et al. 2017), *not* a Hinton-style KD temperature (the student logits are untempered and there is no T^2 term). The justification is Menon et al. (2021): distillation helps insofar as the teacher approximates the Bayes class-probability, which calibration improves. **The paper must cite Menon, not Hinton, for this**

formula. SegFormer distils online at $\alpha = 0.6$; YOLO distils *offline* via a teacher-fused pseudo-mask at $\alpha = 0.4$ (below 0.5 so it is non-degenerate), because Ultralytics’ native trainer consumes only a binary mask.

4.3 Training configuration and a known confound

YOLO is trained with the **native Ultralytics loop** (mosaic, EMA, letterbox, its own cosine LR, auto-batch), not forced onto the transformer recipe—a CNN and a transformer do not share an optimal LR, and native training is what reproduces YOLO’s ≈ 0.83 median. SegFormer uses a per-model batch (largest that fits, capped at 64). The realised configuration, confirmed from the run’s `config.json`, is in Table 1. Because batch is set by model size, **the B2 teacher runs ≈ 2100 optimiser steps while each B0 student runs ≈ 1000** —the teacher receives roughly twice the gradient updates at the same LR. This is an unequalised confound in the teacher–student comparison and is stated as a limitation (Section 12).

Table 1: Realised training configuration (100 epochs, 3 seeds each). B2 confirmed from `config.json`; B0 as configured; YOLO uses Ultralytics auto-batch.

Model	Effective batch	Steps/epoch	Total steps
SegFormer-B2 teacher	32	21	2100
SegFormer-B0 direct	64	10	≈ 1000
SegFormer-B0 distilled	64	10	≈ 1000
YOLO26n direct/distilled	Ultralytics auto (−1)	—	native

4.4 Evaluation and statistics

Everything is scored on a 640×640 grid. YOLO is evaluated *two ways*: (a) **native argmax** (its home turf, reproducing ≈ 0.83 median), and (b) **custom** /255 (the same 640 geometry SegFormer sees, threshold swept on validation). The primary safety metric is the **complete-miss rate**: the fraction of images that contain a bruise but receive a wholly empty predicted mask. Because the 185 images come from only 28 subjects, all confidence intervals are **subject-level cluster bootstraps** ($B = 4000$; resample subjects, not images), and model-vs-model contrasts are **paired** (the same resample scores both models—correct for a shared test set and $\approx 2\times$ tighter than unpaired). We report $P(\Delta > 0)$ alongside each interval to distinguish an underpowered null from a genuine coin-flip.

5 Results: Accuracy and Efficiency

5.1 Headline accuracy

Table 2 gives the three-seed mean $\pm$ std; Table 3 gives the best seed selected on validation and reported on test.

The SegFormer models cluster tightly (0.760–0.765 mean, ≈ 0.811 median). With batch equalised at the level of *family* the teacher leads; an earlier run in which a student appeared to beat the teacher was traced to the batch/step confound (Section 12) and does not survive. Distillation reads

Table 2: Test accuracy, mean $\pm$ std over seeds 0/1/2 (185 images). YOLO shown for both evaluation paths.

Variant	Mean Dice	Median Dice	Complete-miss %
SegFormer-B2 teacher	0.765 $\pm$ 0.004	0.811	0.00
SegFormer-B0 distilled	0.764 $\pm$ 0.005	0.811	0.18 $\pm$ 0.31
SegFormer-B0 direct	0.760 $\pm$ 0.006	0.811	0.54 $\pm$ 0.00
YOLO26n direct (native argmax)	0.691 $\pm$ 0.020	0.802	7.57 $\pm$ 1.08
YOLO26n direct (custom /255)	0.693 $\pm$ 0.026	0.783	4.68 $\pm$ 2.72
YOLO26n distilled (native argmax)	0.668 $\pm$ 0.065	0.766	7.57 $\pm$ 4.80
YOLO26n distilled (custom /255)	0.624 $\pm$ 0.091	0.713	3.42 $\pm$ 2.44

Table 3: Best seed (selected on validation Dice), reported on test.

Model	Seed	Val Dice	Test mean	Test median	Miss %
SegFormer-B2 teacher	0	0.772	0.769	0.819	0.00
SegFormer-B0 distilled	0	0.766	0.768	0.817	0.00
SegFormer-B0 direct	0	0.756	0.766	0.813	0.54
YOLO26n direct (native)	2	0.731	0.702	0.806	6.49
YOLO26n distilled (native)	0	0.743	0.726	0.801	2.16

as a *variance and miss stabiliser* rather than a Dice gain: the distilled B0 has the lowest miss rate of the students and a tighter seed spread.

5.2 Efficiency

Table 4 reports single-frame latency (640 tensor on GPU $\rightarrow$ mask on GPU, A100, seed 0). YOLO is $\approx 4\times$ faster than SegFormer-B0 and $\approx 15\times$ faster than the B2 teacher, at 1.63 M parameters—but, as the miss rates show, it pays for that speed in silent failures.

Table 4: Inference latency and size (A100, single frame, seed 0).

Model	Median (ms)	p95 (ms)	FPS	Params (M)	Peak act. (MB)
SegFormer-B2 teacher	33.7	34.9	29.7	27.35	1779
SegFormer-B0 direct	16.7	16.9	59.9	3.71	1204
SegFormer-B0 distilled	16.6	17.1	60.1	3.71	1204
YOLO26n direct	8.2	8.4	122.2	1.63	967
YOLO26n distilled	8.2	8.5	121.7	1.63	967

6 Results: Which Differences Are Real?

Table 5 gives the paired subject-level contrasts. Only two clear zero: **SegFormer beats YOLO on Dice** (+0.066, 95% CI [0.011, 0.121]), and the **miss-rate gap** between YOLO-direct and the distilled B0 (+6.49 points, 95% CI [2.39, 11.63]). The two intra-family Dice contrasts that the study

was originally designed around—distillation and student–teacher—are both unresolvable, and their $P(\Delta > 0)$ values (0.58 and 0.44) confirm they are genuine coin-flips at this sample size, not merely “not yet significant.”

Table 5: Paired subject-level cluster-bootstrap contrasts (best seed, $B = 4000$). A contrast is resolvable only if its interval excludes zero. All rows are Dice except the last, which is complete-miss rate in percentage points.

Contrast	Δ	95% CI	$P(\Delta > 0)$	Verdict
Distillation (B0-dist – B0-direct)	+0.002	[−0.009, +0.013]	0.58	n.s.
Student – teacher (B0-dist – B2)	−0.001	[−0.020, +0.018]	0.44	n.s.
SegFormer – YOLO (B0-dist – YOLO-direct)	+0.066	[+0.011, +0.121]	0.99	significant
YOLO distilled – direct	+0.024	[−0.014, +0.059]	0.90	n.s.
MISS%: YOLO-direct – B0-dist	+6.49	[+2.39, +11.63]	1.00	significant

7 Results: Failure Modes (the axis that matters)

For an injury-documentation tool the decisive failure is a *complete miss*: a real bruise, and the model returns nothing. A loose outline is correctable; a blank mask is silence about a present injury, and median Dice cannot see the difference. Table 6 shows the miss rates with cluster-bootstrap intervals; note that the model with the *highest* median Dice (YOLO-direct native, 0.806) also has the *highest* miss rate—best Dice is not the safest model.

Table 6: Complete-miss rate (best seed) with subject-bootstrap 95% CI.

Model	Miss %	95% CI	Median Dice
SegFormer-B2 teacher	0.00	[0.00, 0.00]	0.819
SegFormer-B0 distilled	0.00	[0.00, 0.00]	0.817
SegFormer-B0 direct	0.54	[0.00, 1.73]	0.813
YOLO26n distilled	2.16	[0.52, 4.21]	0.801
YOLO26n direct	6.49	[2.31, 11.63]	0.806

Seed instability of YOLO misses. The miss failure is not only larger for YOLO, it is unstable across seeds. Per-seed complete-miss counts (native argmax, out of 185): YOLO-direct **16** / **14** / **12**; YOLO-distilled **4** / **17** / **21**. By contrast the SegFormer teacher never blanks (0/0/0), B0-direct blanks 1/1/1, and B0-distilled 0/0/1. YOLO’s coarse stride-8 head makes its recall brittle in a way post-processing cannot repair.

8 Results: The Annotation Ceiling (the headline)

The most important result in the project concerns the labels, not the models. Measuring per-image Dice *between* the three human experts on the same 185 images (pure mask arithmetic, no model) gives Table 7.

Table 7: Inter-annotator agreement (mean Dice, 185 images).

Comparison	Mean Dice
Paul vs. Gbarimah	0.581
Paul vs. Erik	0.581
Gbarimah vs. Erik	0.755
Average human–human	0.639
Paul vs. majority	0.700
Gbarimah vs. majority	0.873
Erik vs. majority	0.866

Three trained experts agree with one another at only **0.639** mean Dice. Every model scores *above* that against the consensus (0.702–0.769). Note that Paul—whose masks we *train on*—is the outlier: he matches the majority least (0.700) of the three, because the majority is effectively “what Gbarimah and Erik agreed on” (their mutual agreement, 0.755, is far higher than either has with Paul).

8.1 The model beats the annotator it learned from

This sets up the headline. Each model is trained only on Paul’s masks yet scored against the three-expert majority; does it agree with the consensus better than Paul does? Table 8 answers yes, significantly, for all three SegFormer models.

Table 8: Model vs. Paul against the same majority target (paired subject bootstrap, $B = 4000$).

Model	Model vs. maj.	Δ (model – Paul)	95% CI	$P(\Delta > 0)$
SegFormer-B2 teacher	0.769	+0.069	[+0.010, +0.124]	0.99
SegFormer-B0 direct	0.766	+0.067	[+0.005, +0.127]	0.98
SegFormer-B0 distilled	0.768	+0.068	[+0.003, +0.133]	0.98
YOLO26n direct	0.702	+0.002	[−0.049, +0.060]	0.51
YOLO26n distilled	0.726	+0.026	[−0.020, +0.075]	0.86

A model trained on a single annotator agrees with the three-expert consensus significantly better than that annotator does (SegFormer: $\Delta \approx +0.07$, $P \approx 0.98$). Training averages away the annotator’s idiosyncrasies and keeps what generalises. For scale, this effect is $\approx 35\times$ the distillation effect and, unlike it, is significant. It also explains the null Dice contrasts: model-vs-model gaps of 0.002 are being measured against a target carrying ≈ 0.36 of annotation noise ($1 - 0.639$). The models are **label-limited**, **not capacity-limited**.

9 Results: Fairness Across Skin Tone (exploratory)

Because this is a forensic tool, subgroup behaviour across skin tone is a primary concern, not an ablation. We stress up front that with only ≈ 9 –17 subjects per ITA group these results are **exploratory**: they indicate direction, not a proven disparity.

9.1 Group-level gaps

Table 9 (best-seed, computed with the same routine that produced the three-seed `fairness_stats.csv`) reports the omnibus Kruskal–Wallis test and the best–worst-group median-Dice gap. The SegFormer models show small gaps and *zero* misses in every group. The YOLO models’ worst group is consistently **Light (II–III)**, with a within-group complete-miss rate up to 15% (native direct).

Table 9: Best-seed fairness across ITA groups. “KW p ” is the omnibus Kruskal–Wallis p -value; gap is best–worst group median Dice; max miss gap is the largest between-group difference in complete-miss rate.

Variant	KW p	Dice gap	Worst group	Max miss gap
SegFormer-B2 teacher	0.028	0.094	Tan (IV)	0.00
SegFormer-B0 direct	0.709	0.064	Dark (VI)	0.03
SegFormer-B0 distilled	0.749	0.043	Dark (VI)	0.00
YOLO26n direct (native)	0.212	0.131	Light (II–III)	0.15
YOLO26n direct (custom)	0.185	0.120	Light (II–III)	0.13
YOLO26n distilled (native)	0.063	0.075	Light (II–III)	0.08
YOLO26n distilled (custom)	0.094	0.128	Dark (VI)	0.03

The B2 teacher’s omnibus test is nominally “significant” ($p = 0.028$), but with *zero* misses in every group it is a Dice-distribution effect, not a safety disparity—an important distinction not to overstate. Figure 1 shows the complete-miss rate by skin tone across all variants (three-seed).

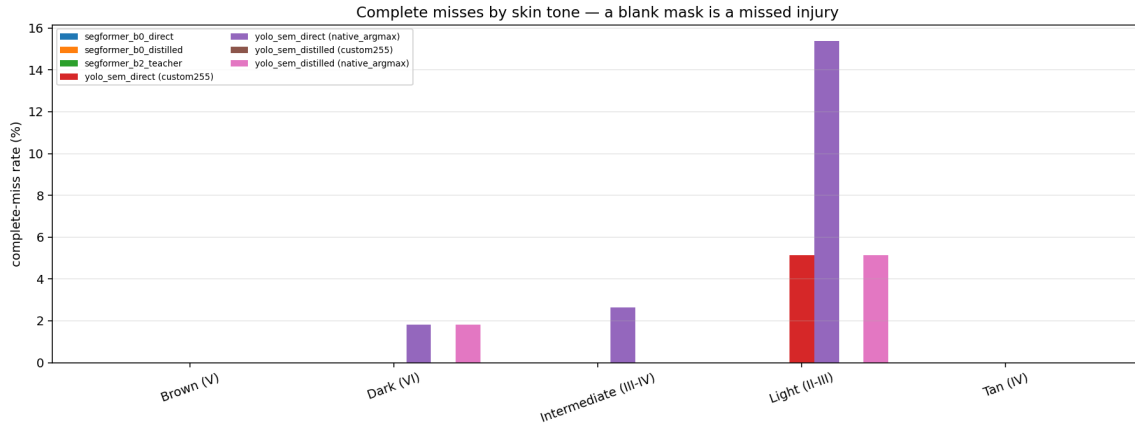


Figure 1: Complete-miss rate by ITA skin-tone group (three-seed). SegFormer variants sit at zero across groups; the YOLO variants concentrate their misses on the lightest-skin group.

10 Results: Bruise Size, and the Size–Fairness Confound

Small bruises are intrinsically harder—a few pixels of boundary error cost proportionally more Dice on a small target, and misses concentrate on the smallest lesions. This matters for Section 9 because **lesion size is confounded with skin tone in this cohort**. Median bruise size by ITA group: Light (II–III) **8,085** px, Brown 10,961, Intermediate 12,118, Tan 13,550, Dark 13,751. Light-skin bruises are $\approx 33\%$ smaller than the median of the rest (Kruskal–Wallis $H = 21.1$, $p = 0.0003$).

10.1 Does the skin-tone signal survive size?

We test this directly, with **standard errors clustered by subject** (treating 185 images as independent would badly overstate significance for a 9-subject group). Two findings:

- **Stratified.** Within the smallest size tercile of YOLO-direct (where size is roughly held fixed), the light-skin group still fares worse: complete-miss **30%** vs. **9.5%**, mean recall **0.41** vs. **0.69**. The gap does not vanish under size control.
- **Regression.** In $\text{recall} \sim \log_{10}(\text{size}) + \text{light}$ with subject-clustered SEs, the light coefficient is -0.176 ($p = 0.047$) for YOLO-direct and -0.164 ($p = 0.045$) for YOLO-distilled, while $\log(\text{size})$ is itself non-significant—so the light-skin recall deficit is *not only* size. For the binary complete-miss outcome the light term is non-significant once size is in ($p = 0.62$); it is the graded recall, not the blank-mask event, that carries the residual signal. The SegFormer models show no light-skin effect ($p > 0.39$).

Clustering is decisive here: the naive, image-independent version of the same recall regression gives $p = 0.0016$, an order of magnitude smaller than the honest $p = 0.047$. The contrast is itself a clean illustration of benchmark fragility. **Net reading:** the apparent skin-tone gap is partly a size artefact, but a residual light-skin recall deficit for YOLO survives size and clustering at the boundary of significance. At 9 light-skin subjects this is a hypothesis to test on more data, not a proven disparity.

11 Discussion

Three themes recur. **(1) Label-limited, not capacity-limited.** All transformer variants—and the U-Net / DeepLabV3+ CNN baselines (Section 13)—sit within a Dice band (0.745–0.765) narrower than human–human disagreement; seven architectures cluster there, and the binding constraint is annotation noise (Section 8), not model capacity. **(2) Miss rate is the honest axis.** It is the one quantity that separates architectures by more than label noise and the one that matters clinically; SegFormer is essentially miss-free while YOLO is not. **(3) Benchmark fragility.** The headline numbers move more with harness choices—subject vs. image resampling, evaluation path (YOLO native vs. custom), preprocessing, batch/step budget—than with architecture. A study that reports a single accuracy number, treats images as independent, or fixes a threshold at 0.5 would draw materially different (and less defensible) conclusions.

Deployment and licensing. For a non-commercial forensic tool both model families are usable, but the licenses differ: the SegFormer pretrained weights carry an NVIDIA non-commercial license, and Ultralytics/YOLO is AGPL-3.0 (network copyleft; reimplementing the code does not free the weights). On both accuracy-safety (misses) and license risk, SegFormer is the preferable deployment target, with YOLO retained as a fast baseline.

12 Limitations

- **Sample size is a hard ceiling.** $n = 28$ subjects is the maximum for which three-annotator consensus labels exist; the minimum resolvable mean-Dice effect is ≈ 0.04 .

- **Unequalised training budget.** With per-model batch, the B2 teacher trains ≈ 2100 steps vs. ≈ 1000 for each B0 student (Table 1)—a confound in the teacher–student comparison. A batch-matched run (effective batch 8 for all) exists separately and could be used to isolate architecture from budget.
- **Single site, single modality** (white light), single annotator for training (Paul).
- **Fairness is exploratory** (≈ 9 – 17 subjects/group); the size–skin-tone confound is real and only partially resolved.
- **Single-seed inference** for the analysis figures (the best validation seed); three-seed aggregates are reported for the headline tables.

13 External Baselines: U-Net and DeepLabV3+

Two external baselines were trained in `bruise_colab_baselines.ipynb` through the *identical* SegFormer shared recipe (they are plain `nn.Modules`: an ImageNet-pretrained ResNet50 encoder + a randomly-initialised decoder, so the shared recipe is fair), on the same 697/134 split, **seeds 0/1/2**, scored against the same 2-of-3 majority target at 640 with complete-miss rate reported. Results are the three-seed mean $\pm$ std in Table 10. (*nnU-Net v2, the native self-configuring “each-at-its-strongest” datapoint, has not been run yet; it will be added later.*)

Table 10: External baseline results (3 seeds, 185 test images), next to the core SegFormer models for context. All trained with the shared recipe; scored vs the 2-of-3 majority @640.

Model	Params (M)	Mean Dice	Median Dice	Miss %
SegFormer-B2 teacher (<i>core</i>)	27.35	0.765 ± 0.004	0.811	0.00
SegFormer-B0 direct (<i>core</i>)	3.71	0.760 ± 0.006	0.811	0.54
U-Net (ResNet50)	32.52	0.753 ± 0.015	0.826	3.24 ± 0.54
DeepLabV3+ (ResNet50)	26.68	0.745 ± 0.018	0.814	1.80 ± 0.62
nnU-Net v2 (2d)	—	<i>pending</i>	<i>pending</i>	<i>pending</i>

The baselines reinforce the annotation-ceiling thesis. With U-Net and DeepLabV3+ added, **seven architectures now span roughly 0.745–0.765 mean Dice** (SegFormer 0.760–0.765, U-Net 0.753, DeepLabV3+ 0.745; the YOLO family lower at 0.69–0.73). U-Net in particular is statistically indistinguishable from the SegFormer-B0 students: the point gap SegFormer-B0-direct – U-Net is only ≈ 0.007 , and even SegFormer-B2 – DeepLabV3+ is ≈ 0.020 —both at or below the ≈ 0.04 minimum detectable effect at $n = 28$. This is exactly the pattern the label-limited hypothesis predicts: swapping a transformer for a CNN, or U-Net for DeepLabV3+, moves Dice less than the annotation noise. **We therefore do not claim “SegFormer beats U-Net/DeepLabV3+”** on Dice.

Miss rate still orders the architectures the same way. The safety axis is again more informative than Dice: SegFormer (0–0.5%) < DeepLabV3+ (1.80%) < U-Net (3.24%) < YOLO (2–8%). The transformer remains the safest; among the CNN baselines, DeepLabV3+’s ASPP context gives it a lower blank-mask rate than U-Net.

Fairness (baselines). Neither baseline shows a significant ITA effect (Kruskal–Wallis $p = 0.22$ U-Net, $p = 0.11$ DeepLabV3+; best–worst-group median-Dice gap 0.096 and 0.079), consistent with the exploratory-at- $n=28$ reading of Section 9. Per-group medians are in Figure 2.

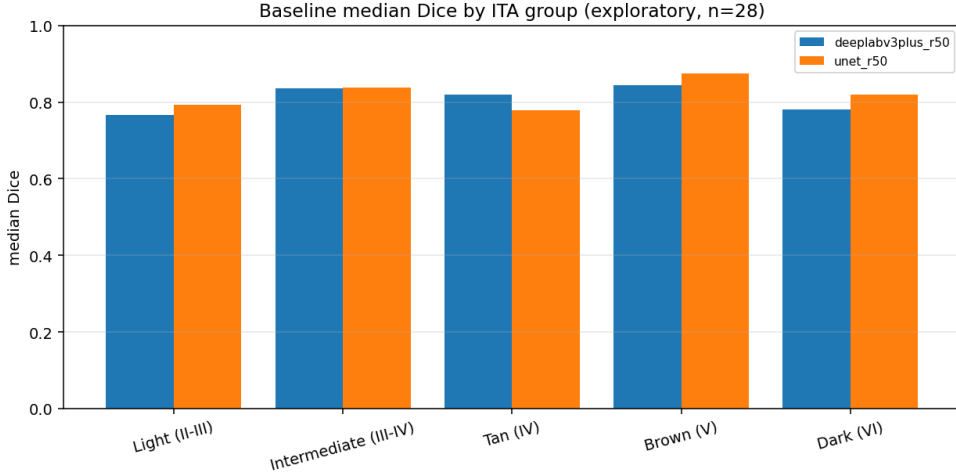


Figure 2: Baseline median Dice by ITA skin-tone group (three-seed; exploratory at $n = 28$).

Best-seed selection (parity with the core best-seed table). To compare the baselines on the *same* policy as Table 3—best seed chosen on *validation* Dice, then scored on test—each baseline’s best of three seeds was selected from its saved validation operating point (`operating_point.json`); results in Table 11. The val-selected baselines rise slightly (best-of-3 vs. mean-of-3) but the ordering is unchanged: both sit below the three SegFormer models on mean Dice and blank more often. Notably U-Net posts the *highest median Dice of any model* (0.833), yet its larger mean–median gap and 3.78% miss rate reflect more complete failures—the same median-is-not-safety lesson as YOLO.

Table 11: Baselines, best seed selected on validation Dice (same policy as Table 3); that seed’s test result reported.

Model	Params (M)	Test mean Dice	Test median Dice	Miss %
DeepLabV3+ (ResNet50)	26.68	0.758	0.818	2.16
U-Net (ResNet50)	32.52	0.757	0.833	3.78

The paired bootstrap is now unblocked. The per-image test scores needed for a paired subject-level cluster-bootstrap of each baseline against the core models (as in Table 5) are available in `runs_baselines/*/test_per_image.csv`; each baseline-vs-core comparison should be reported as Δ mean/median Dice + 95% CI + $P(\Delta > 0)$. Given the point gaps (≈ 0.007 – 0.020 , all at or below the ≈ 0.04 minimum detectable effect at $n = 28$), most are expected to be n.s.

14 Reproducibility and Artifacts

- **Training/eval:** `bruise_colab_final.ipynb` (generator scripts/43_generate_final_notebook.py); five models, three seeds, native YOLO, two YOLO eval paths, ITA fairness. Outputs in `results_final/` and checkpoints in `runs_final/`.
- **Analysis:** `bruise_colab_final_analysis.ipynb` (generator scripts/44_generate_final_analysis_notebook.py); best-seed re-inference, ≈ 25 figures, annotation ceiling, model-vs-Paul, paired contrasts with $P(\Delta > 0)$, ITA fairness, bruise size, and the size-conditioned regression of Section 13–10.
- **Baselines:** `bruise_colab_baselines.ipynb` (generator scripts/45_generate_baselines_notebook.py).
- All numeric results in this report are read directly from the run CSVs (`FINAL_RESULTS.csv`, `*_test_per_seed.csv`, `benchmark_640.csv`, `fairness_*.csv`) and the analysis outputs (`final_analysis_*/`); none are transcribed by hand.

Key references

- Xie et al., *SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers*, NeurIPS 2021.
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- Guo et al., *On Calibration of Modern Neural Networks*, ICML 2017 (temperature scaling).
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- Milletari et al., *V-Net* (Dice loss), 3DV 2016; Drozdal et al. 2016 (Dice+BCE).
- Chardon et al., *Individual Typology Angle*, 1991 (objective skin tone).
- Efron & Tibshirani, *An Introduction to the Bootstrap*, 1993 (cluster/percentile bootstrap).